PRISMSPACE

Project Group 01

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Presentation roadmap

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Introduction

Overview

While modern development tasks span code, browsers, databases, and **APIs** (Application Programming Interfaces), uncoordinated multi-agent workflows suffer from context fragmentation, runaway loops, and tool hallucinations. To address these challenges, PrismSpace introduces a unified developer workspace featuring **multi-agent orchestration**, sandboxed **MCP** (Model Context Protocol) tool execution with strict **ReAct** (Reasoning and Acting) loops, **HITL** (Human-in-the-Loop) safety, and **ORPO** (Odds Ratio Preference Optimization) preference alignment.

Objectives

Core Objectives

The primary objective is to enable structured task execution through **DAG** (Directed Acyclic Graph)-based task decomposition, **MCP** (Model Context Protocol)-enabled multi-agent **ReAct** (Reasoning and Acting) orchestration, and **ORPO** (Odds Ratio Preference Optimization) preference optimization. Additionally, the system ensures reliable and cost-effective operation by incorporating supervised **LLM** (Large Language Model) routing alongside Isolation Forest anomaly detection and human-in-the-loop validation.

Problem Statement

Problem Formulation

Because ungrounded **LLM** (Large Language Model) agents risk hallucinating tool results and executing unsafe actions, the system takes developer task prompts, workspace states, and **MCP** (Model Context Protocol)/**HITL** (Human-in-the-Loop) configurations as its primary **inputs**. Through an orchestration **method** integrating sandboxed **MCP ReAct** (Reasoning and Acting) loops, **ORPO** (Odds Ratio Preference Optimization), supervised **LLM** routing, and Isolation Forest anomaly detection with **HITL** oversight, it converts these tasks into dependable **outputs** consisting of verified artifacts, **ReAct** execution traces, and cost telemetry.

Multi-Objective Reward Generalization in Deep Reinforcement Learning

Citation	Summary	Advantages & Disadvantages
F. Cornalba, C. Disselkamp, D. Scassola, and C. Helf *Neural Computing and Applications* **Year:** 2024 **Publisher:** Springer Open Access	Investigates **Multi-Objective Deep Reinforcement Learning (MODRL)** for optimizing competing operational objectives (e.g., return vs. risk). Introduces reward generalization techniques that train pareto-optimal policies capable of dynamically adapting to shifting stakeholder preferences without retraining from scratch.	**Advantages:** ▶ Provides structured Pareto-optimal policy exploration across conflicting objectives. ▶ Generalizes across parameterized reward weights without full policy retraining. ▶ Directly informs multi-objective routing (latency vs. cost vs. quality) in PrismSpace. **Disadvantages:** ▶ High initial training sample complexity across multi-dimensional reward spaces. ▶ Evaluated primarily on continuous financial time-series data. ▶ Sensitive to hyperparameter tuning in policy entropy regularization.

Average Reward Adjusted Discounted Reinforcement Learning

Citation	Summary	Advantages & Disadvantages
M. Schneckenreither and G. Moser *Neural Computing and Applications* **Year:** 2025 **Publisher:** Springer Open Access	Proposes **Average-Reward Adjusted Discounted RL (Reinforcement Learning) (ARAD-RL)**, bridging the gap between discounted and average reward criteria in infinite-horizon **MDPs** (Markov Decision Processes). Addresses systematic value estimation bias in standard discounted Q-learning by dynamically subtracting the estimated gain rate during Bellman updates.	**Advantages:** ▶ Eliminates infinite-horizon return explosion while preserving policy convergence. ▶ Improves long-term steady-state throughput in continuous scheduling tasks. ▶ Theoretically sound formulation for continuous agent lifecycle optimization. **Disadvantages:** ▶ Requires accurate empirical estimation of average reward baseline. ▶ Slow convergence in highly non-stationary transition environments. ▶ Adds variance to early-stage **TD** (Temporal Difference) error gradient estimates.

Deep Reinforcement Learning for Multi-Agent Coordination

Citation	Summary	Advantages & Disadvantages
K. Aina and S. Ha *Artificial Life and Robotics* **Year:** 2026 **Publisher:** Springer Open Access	Develops decentralized **MARL** (Multi-Agent Reinforcement Learning) algorithms for multi-agent coordination in constrained environments. Employs **spatial-temporal observation encoding** and localized interaction graphs to coordinate joint movements and resolve deadlocks without requiring centralized global state knowledge.	**Advantages:** ▶ Resolves shared-resource contention and deadlocks via decentralized policies. ▶ Spatial-temporal graph attention scales linearly with agent density. ▶ Applicable to concurrent tool-use contention across worker agents. **Disadvantages:** ▶ Tested in physical robotics grids rather than software **API** (Application Programming Interface) environments. ▶ Vulnerable to communication packet dropouts and sensor latency. ▶ Does not account for semantic language-based coordination.

Open-Ended Coordination Using Modular Open Policies

Citation	Summary	Advantages & Disadvantages
D. Rother, J. Pajarinen, J. Peters, and T. H. Weisswange *Autonomous Agents and Multi-Agent Systems* **Year:** 2025 **Publisher:** Springer Open Access	Presents a framework for **open-ended ad hoc multi-agent coordination**. Utilizes **Modular Open Policies (MOP)** where dynamic agent teams dynamically infer unseen teammate skill representations and synthesize blended coordination strategies without requiring pre-coordinated joint training.	**Advantages:** ▶ Enables zero-shot ad hoc coordination with unfamiliar teammate agent policies. ▶ Modular policy decomposition allows plug-and-play agent role assignment. ▶ Directly adopted in PrismSpace for dynamic worker swarm scaling. **Disadvantages:** ▶ Requires comprehensive pre-trained skill policy libraries for base roles. ▶ Evaluated primarily on benchmark games (Overcooked, **LBF** (Level-Based Foraging)). ▶ Action blending may introduce transient sub-optimal execution steps.

Improving Scalability of **MADRL** (Multi-Agent Deep Reinforcement Learning) with Suboptimal Human Knowledge

Citation	Summary	Advantages & Disadvantages
D. Liu, F. Ren, J. Yan, G. Su, W. Gu, and S. Kato *Autonomous Agents and Multi-Agent Systems* **Year:** 2026 **Publisher:** Springer Open Access	Addresses sample inefficiency in large-scale **Multi-Agent DRL** (Deep Reinforcement Learning) by integrating heuristic human guidance into exploration. Proposes an adaptive filtering mechanism that extracts actionable macro-actions from noisy or sub-optimal human demonstration traces, accelerating multi-agent policy learning.	**Advantages:** ▶ Accelerates policy exploration by up to 4x using noisy demonstration priors. ▶ Robust to suboptimal or conflicting human heuristic rules. ▶ Informs PrismSpace's bootstrap initialization from logged developer sessions. **Disadvantages:** ▶ Quality of learned policies depends on initial heuristic coverage. ▶ Requires complex threshold tuning to transition away from demonstrations. ▶ High memory footprint for maintaining dual replay buffers.

ε-Retraining Reinforcement Learning Algorithms

Citation	Summary	Advantages & Disadvantages
L. Marzari, C. Liu, P. L. Donti, and E. Marchesini *Autonomous Agents and Multi-Agent Systems* **Year:** 2026 **Publisher:** Springer Open Access	Investigates bounded policy fine-tuning through ε-**retraining**, preventing catastrophic forgetting when deep **RL** agents adapt to non-stationary environments. Establishes formal Lipschitz bounds on policy drift, ensuring safe continual learning in deployed agent pipelines.	**Advantages:** ▶ Provides mathematical guarantees against catastrophic policy divergence. ▶ Enables continuous telemetry-driven model fine-tuning in production. ▶ Crucial for stability in PrismSpace's ongoing user interaction learning. **Disadvantages:** ▶ Constrained optimization increases computation per gradient step. ▶ Conservative update bounds may slow adaptation to drastic distribution shifts. ▶ Theoretical guarantees rely on local smoothness assumptions.

Deep Multiagent Reinforcement Learning: Challenges and Directions

Citation	Summary	Advantages & Disadvantages
A. Wong, T. Bäck, A. V. Kononova, and A. Plaat *Artificial Intelligence Review* **Year:** 2023 **Publisher:** Springer Open Access	Surveys **Deep Multiagent Reinforcement Learning (MARL)**, formalizing foundational challenges including non-stationarity, multi-agent credit assignment, and combinatorial action explosion. Evaluates **Centralized Training with Decentralized Execution (CTDE)** and communication architectures for modular orchestration.	**Advantages:** ▶ Comprehensive taxonomy of credit assignment and **CTDE** paradigms. ▶ Establishes formal theoretical criteria for communication protocols. ▶ Provides the theoretical foundation for PrismSpace's modular agent architecture. **Disadvantages:** ▶ Survey paper without empirical experimental evaluation. ▶ Focuses on discrete game simulations rather than **API**-driven software tasks. ▶ Does not cover **LLM** reasoning agents or human-in-the-loop approval gates.

HIAT (Human-in-the-Loop with Auxiliary Tasks): Human-in-the-Loop **RL** with Auxiliary Tasks

Citation	Summary	Advantages & Disadvantages
B. Niu, B. Luo, Y. Cui, X. Xu, Y. Zhao, and Y. Feng *Artificial Intelligence Review* **Year:** 2025 **Publisher:** Springer Open Access	Introduces **HIAT (Human-in-the-Loop with Auxiliary Tasks)**, a framework leveraging human intervention signals as auxiliary self-supervised learning objectives. Models human corrections as feedback gradients that shape internal feature representations, minimizing human intervention frequency over time.	**Advantages:** ▶ Maximizes learning efficiency from scarce human intervention signals. ▶ Significantly reduces human cognitive load through auxiliary reward shaping. ▶ Informs PrismSpace's approval gate telemetry and learning from user edits. **Disadvantages:** ▶ Requires consistent human feedback to avoid biased auxiliary representations. ▶ Intervention timing latency can cause temporal mis-alignment in **RL** updates. ▶ Higher computational overhead during auxiliary task representation training.

ORPO (Odds Ratio Preference Optimization): Monolithic Preference Optimization without Reference Model

Citation	Summary	Advantages & Disadvantages
J. Hong, N. Lee, and J. Thorne ***EMNLP*** / *arXiv:2403.07691* **Year:** 2024 **Area: LLM** Preference Alignment	**Core Idea (Single-Phase Alignment):** Instead of multi-step **RLHF**, **ORPO** teaches desired answers while penalizing bad answers in one unified equation: $$\mathcal{L}_{\text{ORPO}} = \mathcal{L}_{\text{SFT}} + \lambda \cdot \mathcal{L}_{\text{OR}}$$ $$\mathcal{L}_{\text{OR}} = -\log \sigma \left(\log \frac{\text{odds}(y_w \mid x)}{\text{odds}(y_l \mid x)} \right)$$ **Key Terms:** ▶ $\mathcal{L}_{\text{SFT}}$: Base supervised fine-tuning loss (correct code/syntax). ▶ $\mathcal{L}_{\text{OR}}, \lambda$: Odds ratio penalty term and penalty weight. ▶ x, y_w, y_l: Prompt, winning action path, and losing action path.	**Advantages:** ▶ Reference-model-free: eliminates maintaining frozen π_{ref} in memory. ▶ Monolithic single-phase training avoids unstable **PPO** policy optimization. ▶ Directly aligns PrismSpace agent trajectories on paired feedback (*chosen* vs. *rejected*). **Disadvantages:** ▶ Requires high-quality pairwise preference datasets (y_w, y_l). ▶ Sensitive to λ hyperparameter balance.

Reinforcement Online Learning to Rank with Unbiased Reward Shaping

Citation	Summary	Advantages & Disadvantages
S. Zhuang, Z. Qiao, and G. Zuccon *Information Retrieval Journal* **Year:** 2023 **Publisher:** Springer Open Access	Develops an **Online Learning to Rank (OLTR)** framework using counterfactual reinforcement learning. Employs **inverse propensity weighting (IPW)** and unbiased reward shaping to correct for position, presentation, and click bias in user interaction logs during continuous ranker optimization.	**Advantages:** ▶ Guarantees unbiased gradient estimation from observational implicit feedback. ▶ Operates entirely online without requiring manual document relevance labels. ▶ Directly informs PrismSpace's context retrieval ranking feedback loop. **Disadvantages:** ▶ High propensity score variance can destabilize gradient updates. ▶ Requires large volumes of query-click logs to converge. ▶ Assumes position-based user click models that may not hold in code search.

Knowledge Learning for Securing Workflow Scheduling in **MEC** (Mobile Edge Computing)

Citation	Summary	Advantages & Disadvantages
T. M. Ghazal, A. E. Awouda, M. K. Hasan, A. H. A. Rahman, S. Islam, R. A. Saeed, H. Elshafie, A. Iqbal, and A. H. Hussein *Mobile Networks and Applications* **Year:** 2025 **Publisher:** Springer Open Access	Proposes a **Knowledge-Guided Reinforcement Learning (KG-RL)** scheduler for complex workflow **DAGs** in distributed edge-cloud environments. Integrates security constraint validation directly into the state transition model, preventing insecure task placement while optimizing makespan.	**Advantages:** ▶ Enforces strict security boundary constraints during automated task scheduling. ▶ Effectively handles multi-dependency Directed Acyclic Graph (**DAG**) workflows. ▶ Informs PrismSpace's sandboxed tool execution and dependency resolution. **Disadvantages:** ▶ Constraint checking introduces scheduling latency for massive **DAGs**. ▶ Assumes static security classification matrices across network nodes. ▶ Limited evaluation on non-deterministic tool failure scenarios.

Energy-Aware Cooperative Multi-Fitness Evolutionary Workflow Scheduling

Citation	Summary	Advantages & Disadvantages
P. Barredo and J. Puente *Natural Computing* **Year:** 2025 **Publisher:** Springer Open Access	Presents a **Cooperative Multi-Fitness Evolutionary Algorithm (CMFEA)** for multi-objective workflow scheduling. Optimizes trade-offs between execution time, monetary compute cost, and energy consumption using island-model cooperative coevolution across heterogeneous worker pools.	**Advantages:** ▶ Pareto-optimal trade-offs across makespan, financial cost, and energy. ▶ Cooperative sub-population decomposition prevents premature local convergence. ▶ Provides heuristics for PrismSpace's multi-agent model provider selection. **Disadvantages:** ▶ Evolutionary search has higher runtime latency than single-pass ML heuristics. ▶ Cannot adapt instantaneously to real-time **API** rate limit drops. ▶ Requires parameter tuning across mutation, crossover, and migration rates.

Optimized Task Scheduling Using Hybrid **DL** (Deep Learning) and Metaheuristics

Citation	Summary	Advantages & Disadvantages
A. Heirati, M. Fartash, and M. Khalily-Dermany *Neural Processing Letters* **Year:** 2026 **Publisher:** Springer Open Access	Combines **Deep Learning (LSTM (Long Short-Term Memory))** workload prediction with metaheuristic optimization (Harris Hawks Optimization) for proactive workflow dispatching. Predicts task execution durations and memory bottlenecks in advance to prevent node overloading and minimize queue waiting times.	**Advantages:** ▶ Proactive workload forecasting prevents sudden compute node thrashing. ▶ Hybrid pipeline balances search exploration with predictive dispatching. ▶ Relevant for predicting token latency and cost before agent invocation. **Disadvantages:** ▶ Two-stage pipeline introduces inference overhead before scheduling decisions. ▶ **LSTM** prediction accuracy degrades under sudden irregular workload spikes. ▶ Complex offline model retraining requirements.

Cost-Aware Workflow Offloading in Edge-Cloud Computing via **GA** (Genetic Algorithm)

Citation	Summary	Advantages & Disadvantages
S. Abdi, M. Ashjaei, and S. Mubeen *The Journal of Supercomputing* **Year:** 2024 **Publisher:** Springer Open Access	Proposes a **Genetic Algorithm-based workflow offloading** framework considering heterogeneous network bandwidth, deadline constraints, and monetary costs. Implements chromosome repair operators to preserve **DAG** precedence validity throughout mutation and crossover.	**Advantages:** ▶ Guarantees valid **DAG** dependency structures via specialized repair operators. ▶ Explicitly accounts for network transfer latency between pipeline stages. ▶ Informs subtask payload transfer between worker agents in PrismSpace. **Disadvantages:** ▶ Computationally heavy for interactive real-time user query response. ▶ Assumes deterministic network transfer bandwidth in cloud environments. ▶ Does not model dynamic agent failures or retries.

Dynamic Role Discovery and Assignment in Multi-Agent Task Decomposition

Citation	Summary	Advantages & Disadvantages
Y. Xia, J. Zhu, and L. Zhu *Complex & Intelligent Systems* **Year:** 2023 **Publisher:** Springer Open Access	Proposes **dynamic role discovery and assignment** that decomposes complex high-level objectives into lower-level subtasks. Uses learned persona embeddings and mutual information to adaptively allocate subtasks according to situational agent capabilities, eliminating rigid static assignments.	**Advantages:** ▶ Enables flexible subtask allocation without hardcoded role hierarchies. ▶ Hierarchical decomposition prevents single-agent cognitive context overload. ▶ Serves as core theoretical basis for PrismSpace's dynamic agent routing. **Disadvantages:** ▶ Role discovery clustering introduces upfront computational overhead. ▶ Performance depends on representation vector accuracy for subtasks. ▶ Evaluated primarily on synthetic collaborative coordination benchmarks.

Decision Variable Contribution Based Adaptive Multi-Objective Scheduling

Citation	Summary	Advantages & Disadvantages
J. Li, L. Xing, W. Zhong, Z. Cai, and F. Hou *Complex & Intelligent Systems* **Year:** 2023 **Publisher:** Springer Open Access	Introduces a **Decision Variable Contribution (DVC)** adaptive mechanism for evolutionary multi-objective workflow scheduling. Evaluates the differential sensitivity of individual decision variables to focus evolutionary search operators on high-impact scheduling bottlenecks.	**Advantages:** ▶ Accelerates Pareto-front convergence by prioritizing high-variance variables. ▶ Dynamically adapts search intensity across workflow bottleneck stages. ▶ Useful for pinpointing high-latency critical path tools in agent workflows. **Disadvantages:** ▶ Sensitivity calculation adds per-generation matrix computation cost. ▶ Can over-concentrate on dominant objectives at the expense of diversity. ▶ Requires calibrated normalization across widely different metric units.

GHQ: Grouped Hybrid Q-Learning for Heterogeneous Multi-Agent RL

Citation	Summary	Advantages & Disadvantages
X. Yu, Y. Lin, X. Wang, S. Han, and K. Lv *Complex & Intelligent Systems* **Year:** 2024 **Publisher:** Springer Open Access	Develops **Grouped Hybrid Q-learning (GHQ)** for cooperative heterogeneous multi-agent systems. Clusters agents with similar capabilities into functional cohorts that share value approximations while maintaining decoupled action execution heads for role specialization.	**Advantages:** ▶ Scales value estimation across large heterogeneous swarms via cohort grouping. ▶ Reduces joint action space dimensionality from exponential to block-linear. ▶ Directly applicable to categorizing **LLM** worker agent skill specializations. **Disadvantages:** ▶ Cohort clustering boundaries can be rigid if agent capabilities evolve. ▶ Intra-group credit assignment remains non-trivial in dense interactions. ▶ Requires domain knowledge to establish initial grouping heuristics.

Decentralized Multi-Agent Cooperation via Adaptive Partner Modeling

Citation	Summary	Advantages & Disadvantages
C. Xu, J. Wang, X. Zhu, Y. Yue, W. Zhou, Z. Liang, and D. Wojtczak *Complex & Intelligent Systems* **Year:** 2024 **Publisher:** Springer Open Access	Tackles environmental non-stationarity in decentralized multi-agent cooperation using **Adaptive Partner Modeling (APM)**. Instead of relying on expensive continuous inter-agent broadcast communication, agents infer partner policies and intended actions from local observation histories.	**Advantages:** ▶ Minimizes inter-agent communication bandwidth to near zero during execution. ▶ Stabilizes decentralized joint execution despite teammate policy shifts. ▶ Directly adopted in PrismSpace for zero-communication agent coordination. **Disadvantages:** ▶ Partner inference error propagates if historical observations are noisy. ▶ Inference networks introduce local computational and memory overhead. ▶ Assumes partners follow relatively stationary parameterized policy spaces.

A Responsible AI Framework: Pipeline Contextualisation

Citation	Summary	Advantages & Disadvantages
E. Vyhmeister, G. G. Castañé, P.-O. Östberg, and S. Thevenin *AI and Ethics* **Year:** 2023 **Publisher:** Springer Open Access	Formulates an operational **Responsible AI Framework** providing pipeline contextualisation across the full model lifecycle. Specifies auditable checkpoints, data lineage tracking, and bias mitigation protocols for autonomous AI decision-making systems.	**Advantages:** ▶ Holistic governance framework linking technical architecture to ethical safety. ▶ Defines concrete telemetry points for data provenance and action auditing. ▶ Guides PrismSpace's execution audit logging and human inspection UI. **Disadvantages:** ▶ Qualitative framework that requires custom translation into concrete code gates. ▶ Audit logging introduces storage overhead and slight execution latency. ▶ Does not provide automated formal verification of **LLM** reasoning traces.

Meaningful Human Control: Actionable Properties for AI Development

Citation	Summary	Advantages & Disadvantages
L. Cavalcante Siebert, M. L. Lupetti, E. Aizenberg, N. Beckers, A. Zgonnikov, H. Veluwenkamp, D. Abbink, E. Giaccardi, G.-J. Houben, C. M. Jonker, J. van den Hoven, D. Forster, and R. L. Lagendijk *AI and Ethics* **Year:** 2023 **Publisher:** Springer Open Access	Formalizes the philosophical concept of **Meaningful Human Control (MHC)** into actionable software design properties: **Tracking** (system responsive to human moral reasons) and **Tracing** (system actions causally attributable to accountable human supervisors).	**Advantages:** ▶ Translates abstract governance principles into rigorous software specifications. ▶ Establishes tracking and tracing requirements for autonomous tool execution. ▶ Directly guides the design of PrismSpace's Human-in-the-Loop (**HITL**) approval gates. **Disadvantages:** ▶ High human intervention frequency can degrade automated agent throughput. ▶ Defining actionable reason boundaries requires careful domain calibration. ▶ Assumes human supervisor maintains full comprehension of complex agent plans.

CPN-HRL (Cloud-Private-Network Hierarchical Reinforcement Learning): Hierarchical **DRL** for Priority-Aware Task Scheduling

Citation	Summary	Advantages & Disadvantages
Q. Yue, L. Tian, X. Feng, Z. Yuan, S. Wei, J. Xia, B. Chen, X. Song, and Y. Yao *Journal of King Saud University CIS* **Year:** 2026 **Publisher:** Springer Open Access	Presents **CPN-HRL**, a Hierarchical Reinforcement Learning framework for priority-aware task scheduling in Cloud-Edge-End networks. Uses a macro-policy to partition task **DAGs** by priority and latency budgets, while micro-policies schedule subtasks onto optimal compute nodes.	**Advantages:** ▶ Hierarchical policy decomposition effectively handles complex task **DAGs**. ▶ Guarantees priority-aware quality of service under tight latency deadlines. ▶ Serves as primary reference for PrismSpace's **DAG** workflow scheduling engine. **Disadvantages:** ▶ Multi-tier policy architecture requires synchronous training coordination. ▶ Edge node failure requires immediate micro-policy replanning. ▶ Evaluated in simulated telecom cloud-edge topologies.

Automatic Discovery and Description of Human Planning Strategies

Citation	Summary	Advantages & Disadvantages
J. Skirzyński, Y. Jain, and F. Lieder *Behavior Research Methods* **Year:** 2023 **Publisher:** Springer Open Access	Develops an algorithmic framework for **automatically discovering and formalizing planning strategies** from human behavior traces. Models task planning as resource-rational search over Directed Graphical Models, producing interpretable decision trees that characterize human heuristic planning.	**Advantages:** ▶ Extracts interpretable, human-like heuristic planning strategies automatically. ▶ Quantifies trade-offs between cognitive search effort and decision quality. ▶ Informs PrismSpace's hierarchical prompt decomposition and agent planning. **Disadvantages:** ▶ Search space for strategy discovery grows exponentially with problem depth. ▶ Human behavioral experiments conducted in stylized cognitive testbeds. ▶ Static decision trees struggle to capture fluid **LLM** reasoning trajectories.

Data-Intensive Workflow Scheduling Based on **DRL** in Multi-Clouds

Citation	Summary	Advantages & Disadvantages
S. Zhang, Z. Zhao, C. Liu, and S. Qin *Journal of Cloud Computing* **Year:** 2023 **Publisher:** Springer Open Access	Proposes a **Deep Q-Network (DQN)** scheduler optimized for data-intensive workflows across multi-cloud environments. Explicitly models dataset placement and cross-datacenter transfer latency to minimize both workflow completion time and data egress billing.	**Advantages:** ▶ Optimizes data locality to prevent massive cross-cloud egress overhead. ▶ Adaptive reward function balances monetary cost and makespan constraints. ▶ Directly relevant to caching and embedding retrieval across distributed agents. **Disadvantages:** ▶ Assumes deterministic file sizes and network bandwidth availability. ▶ Q-network value overestimation requires double-**DQN** correction. ▶ State representation size scales with the total number of cloud datacenters.

RL-Based Task Scheduling for Sustainable Federated Cloud Computing

Citation	Summary	Advantages & Disadvantages
Z. Wang, S. Chen, L. Bai, J. Gao, J. Tao, R. R. Bond, and M. D. Mulvenna *Journal of Cloud Computing* **Year:** 2023 **Publisher:** Springer Open Access	Develops an **Actor-Critic RL scheduling framework** for federated cloud datacenters that incorporates renewable energy availability into scheduling policies. Dynamically routes compute tasks to datacenters with green energy surpluses without violating user SLA deadlines.	**Advantages:** ▶ Incorporates environmental sustainability and carbon metrics into scheduling. ▶ Actor-Critic policy gradient handles continuous action spaces effectively. ▶ Provides conceptual foundation for multi-attribute provider routing. **Disadvantages:** ▶ Requires real-time telemetry from green energy generation sources. ▶ Trade-off between carbon footprint and network transmission latency. ▶ Sensitive to reward scaling factors between SLA penalties and green rewards.

Hybrid Dolphin Swarm Sparrow Search Multi-Objective Workflow Scheduling

Citation	Summary	Advantages & Disadvantages
S. Subashree, M. Rajakumaran, G. Pushpa, and S. Senthilkumar *Journal of Cloud Computing* **Year:** 2025 **Publisher:** Springer Open Access	Proposes a hybrid bio-inspired metaheuristic combining **Dolphin Swarm Optimization (DSA)** and **Sparrow Search Algorithm (SSA)** for multi-objective cloud workflow scheduling. Optimizes execution makespan, virtual machine utilization, and monetary execution expenditure.	**Advantages:** ▶ Combines global exploration of DSA with local exploitation of SSA. ▶ Demonstrates superior Pareto hypervolume compared to standard **GA** and **PSO** (Particle Swarm Optimization). ▶ Provides strong comparative baselines for workflow optimization algorithms. **Disadvantages:** ▶ Metaheuristic iteration runtime is too slow for sub-second reactive routing. ▶ Multiple tuning parameters (predation rate, vocalization index) required. ▶ Lacks theoretical convergence rate bounds for stochastic optimization.

Research Gap

What the surveyed methods already do

MARL works address **CTDE**, role grouping, and ad-hoc coordination. **ORPO** aligns **LLM** preferences without reference models, while **HIAT** leverages human interventions as auxiliary **RL** signals. **CPN-HRL** and **DAG** schedulers optimize cloud task allocation, and **MHC** formalizes tracking and tracing properties.

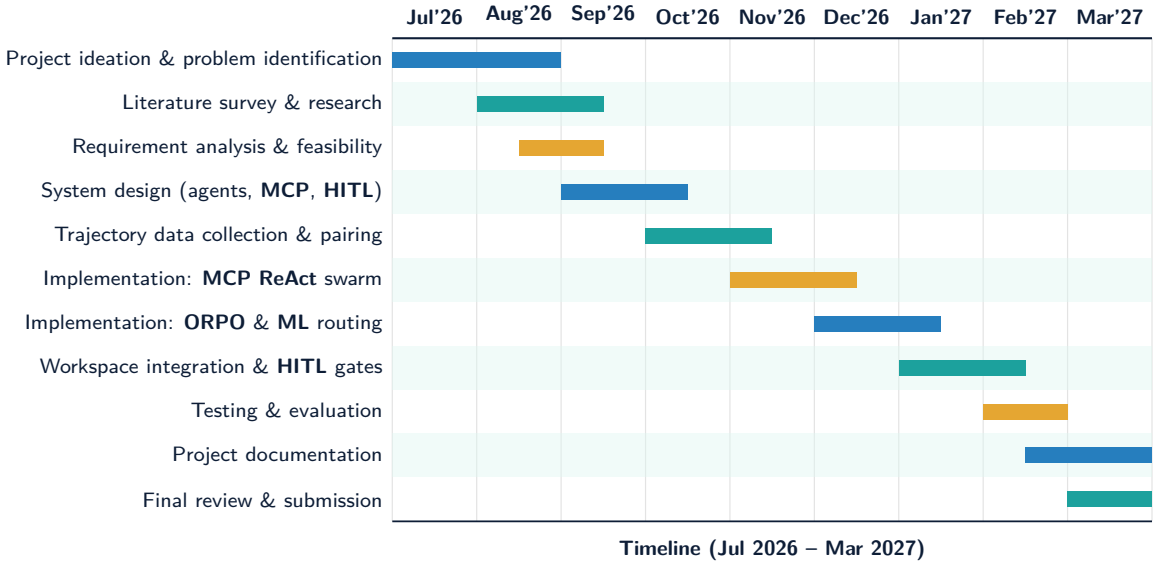
Limitations observed in those papers

Coordination benchmarks focus on gaming (**SMAC**, Overcooked) rather than filesystem/**API** tooling. **ORPO** lacks grounding in **MCP ReAct** tool traces, while **HIAT** targets continuous robotic control instead of code execution. Cloud schedulers assume static **DAGs** or slow metaheuristics ill-suited for real-time **LLM** routing.

Why PrismSpace is required

No reviewed system jointly grounds tool observations, aligns policies on agent trajectories, dynamically routes heterogeneous **LLMs**, and enforces human-in-the-loop safety gates within a unified developer workspace.

Gantt Chart


Jul'26
Aug'26
Sep'26
Oct'26
Nov'26
Dec'26
Jan'27
Feb'27
Mar'27
Project ideation & problem identification
Literature survey & research
Requirement analysis & feasibility
System design (agents, MCP, HITL)
Trajectory data collection & pairing
Implementation: MCP ReAct swarm
Implementation: ORPO & ML routing
Workspace integration & HITL gates
Testing & evaluation
Project documentation
Final review & submission
Timeline (Jul 2026 – Mar 2027)

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Thank You

Questions & Discussion

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